

WHAT CAN THE LOCAL GPS NETWORK TELL US ABOUT THE GLACIAL ISOSTATIC ADJUSTMENT OF LONG ISLAND, NY?

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Abstract

Long Island, NY has a low-topographic relief of large residential areas along the coastline, and a relative subsidence or uplift of areas may elevate the risk of flooding. Utilizing GPS data, the movement of Long Island was analyzed to determine the presence of any glacial isostatic adjustment and positional change. Although Long Island is not in a tectonically active region nor is it directly connected to rebound by changes in mantle viscosity, subsidence or uplift is expected to occur due to glacial isostatic adjustment. Other studies have used models dependent on the last glacial maximum and ice mass change for estimating the severity of global warming and understanding the impact of the relative sea-level change. In our paper, we applied a statistical methodology focused on local changes in elevation and plate positioning. We used an R programming language to statistically analyze the relationship between four GPS sites out of the fourteen on Long Island, specifically their north, east, and up components, to identify any glacial isostatic adjustment that may affect Long Island's topography. A low-pass (KZ) filter has also been applied to remove high-frequencies, allow a higher signal-to-noise ratio, and reveal a long-term trend. The results of this study are consistent with those of previous studies, such as the relative northwestern velocity of the northeastern region of North America, and we have concluded that the four GPS sites with the greatest extended time duration of data show a very prominent and continuous subsidence of the Long Island, NY, with the exception of a short interval of uplift (2013-2015).

Introduction

Objective: observe if glacial melting causes isostatic rebound of Long Island by analyzing GPS data using R programming language and RStudio software

Applying Moving Average:

- A low pass filter may reveal trends within data
- Requires spectral analysis to determine the window size
- Provides best results

Comparing other works:

- Some have used GPSY-OASIS software to analyze velocities of plate motions (Sella et al., 2007), but not specifically for local regions such as Long Island, NY

Methods

- Data: Nevada Geodetic Laboratory's Global GPS Network: Sites NYCI, NYQN, NYRH, and ZNY1 (see Figure 1)
- All figures, data sets cleaning and processing, equations and maps were created through R programming language (Rstudio)
- Created data set of the northward, eastward, and upward movement of all four GPS sites
 - Outliers removed with z-score (-3,3) function in R
 - Spectral analysis applied on raw data (Figure 2)
 - Removed high frequencies accordingly



Figure 1: Map of GPS sites on Long Island, NY (the red represents the GPS sites that were used in our research while the blue are the sites that were not excluded due to the span of their data)

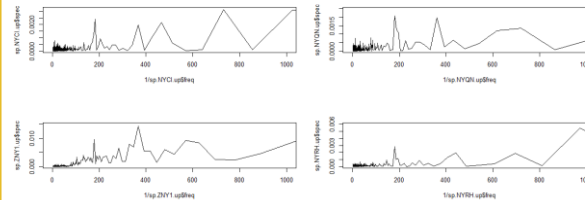


Figure 2: Spectral analysis of the selected GPS sites (horizontal axis is the period in days)

Discussion

- Monitoring of the glacial isostatic adjustment or any vertical displacement is vital to understanding the relative impact of sea level rise. Using GPS coordinates allows us to notice any significant patterns or detect any changes.
- The upward displacement for 2015 (Figure 3) could be caused by an earthquake or GPS modifications.

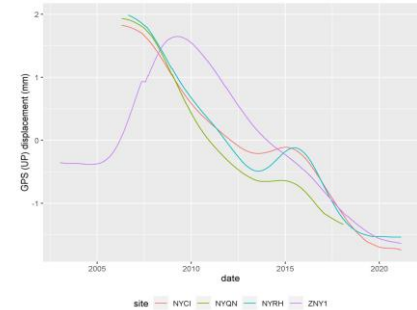


Figure 3: Displacements (in mm; up) from the GPS selected sites NYCI, NYQN, NYRH, and ZNY1 after being averaged by the KZ(730,3) function moving average. Data were standardized to fit at the same scale

Results

- Uniform behavior along the north, east, and vertical components of the GPS sites.
- No presence of uplift related to glacial isostatic rebound.

Conclusion

- Evidence of a consistent downtrend connected to the subsidence of Long Island with no relation to glacial isostatic adjustment.
 - Substantial evidence showing a period of uplift during 3 year period of 2013-2015.

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